UNIVERSITY OF HERTFORDSHIRE

Course - Business Analytics and Consultancy

Module - Computer Simulation for Business (7BSP0413-0206-2023)

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Assignment Type - Team Course Work

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PART A

i) List your variables for the system, ii) put them into linked pairs.

Please provide the list of variables and pairs in your report.

Answers

LIST OF VARIABLES

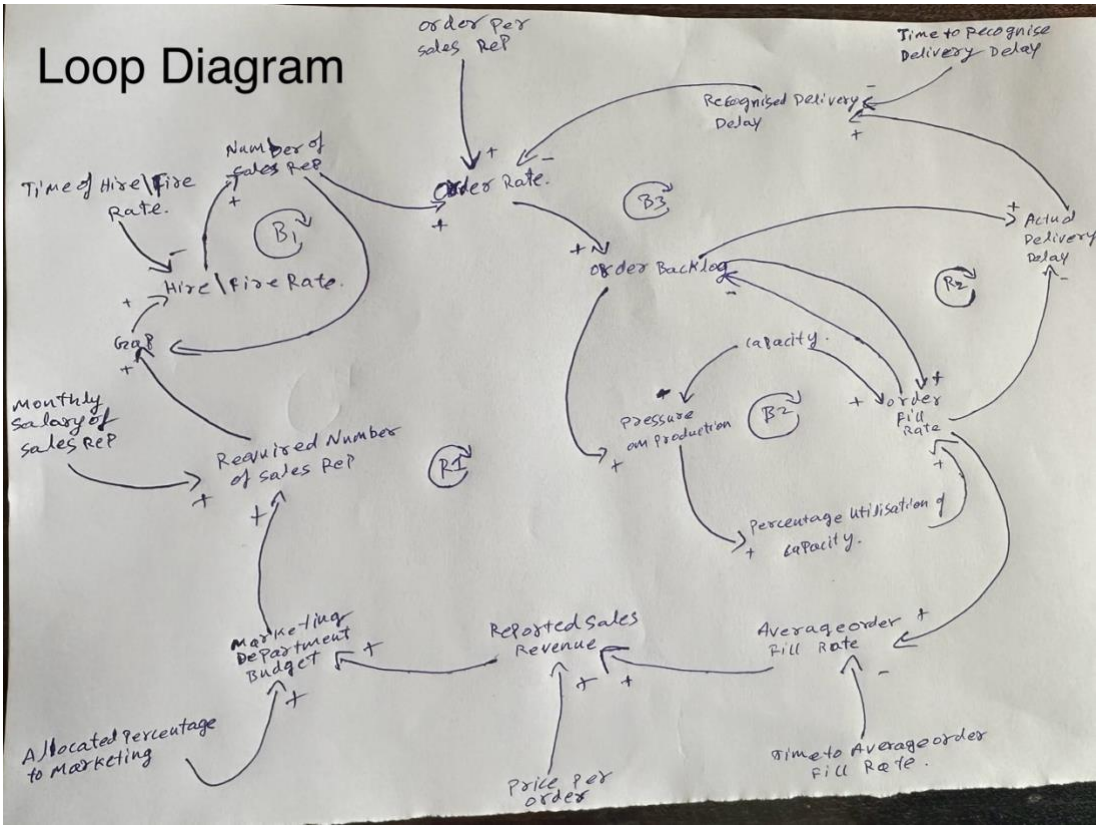
1. Number of sales representatives
2. Order per rep
3. Order capacity
4. Percentage utilization of the capacity
5. Order fill rate
6. Order Backlog
7. pressure on production
8. Reported sales revenue
9. Average order fill rate
10. Time to average order fill rate
11. The budget of marketing department
12. Required number of sales representatives
13. Gap
14. Hire rate/fire rate
15. Time utilized to hire/fire
16. Delivery delay
17. Delivery delay recognized
18. Actual delay in delivery
19. Time to recognize delivery delay
20. Percentage to marketing
21. Order rate (total orders that company receives)
22. Value per order

LINKED PAIRS

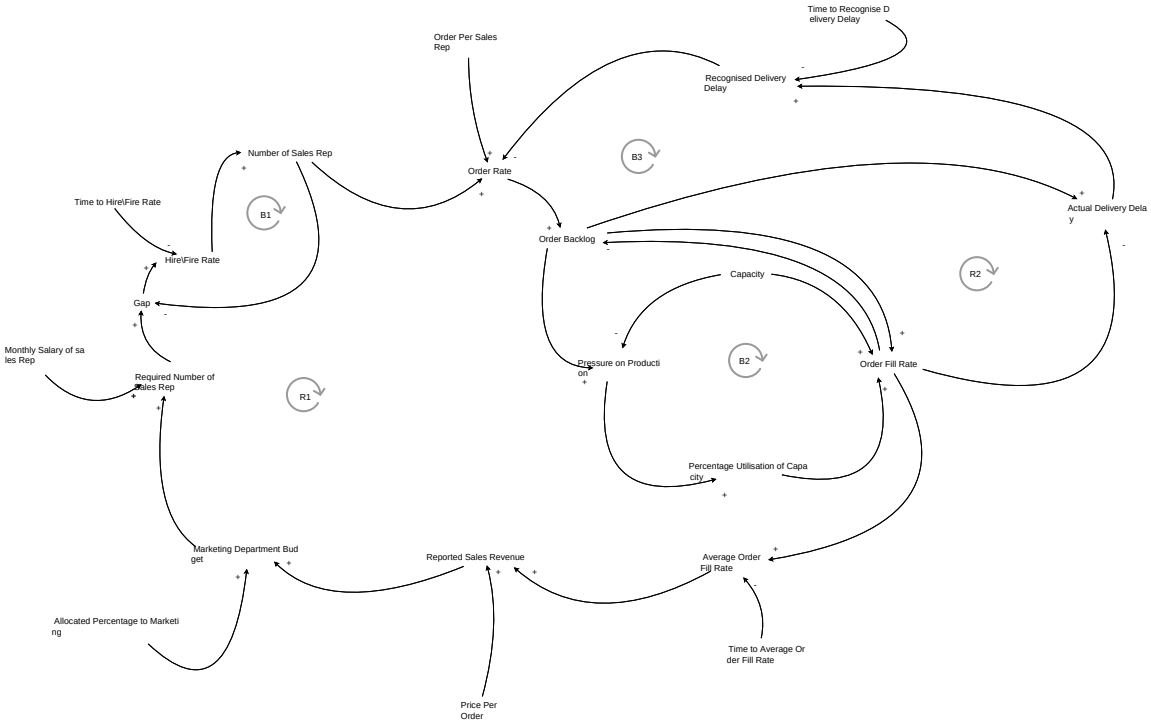
1. Order Rate ---> + Backlog of Orders
2. Backlog of Orders ---> + Pressure On Production
3. Capacity ---> - Pressure On Production
4. Capacity --->+ Order Fill Rate
5. Pressure On Production ---> + Percentage Utilisation
6. Percentage Utilisation ---> + Order Fill Rate
7. Order Fill Rate ---> + Average Order Fill Rate
8. Time To Average Order Fill Rate ---> - Average Order Fill Rate.
9. Average Order Fill Rate ---> + Reported Sales Revenue
10. Value Or Price Per Order ---> + Reported Sales Revenue
11. Reported Sales Revenue ---> + Marketing Budget
12. Percentage To Marketing ---> + Marketing Budget
13. Marketing Budget ---> + Required Number of Representatives
14. Rep Salary ---> - Required Number of Representatives
15. Required Number of Representatives ---> + Gap
16. Number Of Sales Representatives ---> - Gap
17. Gap ---> + Hire/Fire Rate
18. Time To Hire/Fire ---> - Hire/Fire Rate
19. Hire / Fire Rate ---> + Number Of Sales Representatives
20. Number Of Sales Representatives ---> + Order Rate
21. Orders Per Sales Representative ---> + Order Rate
22. Order Fill Rate ---> - Backlog Of Orders
23. Backlog Of Orders ---> + Order Fill Rate
24. Backlog Of Orders ---> + Actual Delivery Delay
25. Order Fill Rate ---> - Actual Delivery Delay
26. Actual Delivery Delay ---> + Recognized Delivery Delay
27. Time To Recognise Delivery Delay ---> - Recognized Delivery Delay
28. Recognised Delivery Delay ---> - Order Rate

ii) Draw a full Causal Loop Diagram for the problem situation or system described below in the case study. Show for each loop whether it is reinforcing or balancing. Do draft or rough versions and please include these as photos, and finally a neat, finished version.

Ans: <u>Draft version</u>



Finished Version



iii) Fully describe each of the main loops, parts or areas of the system, as represented by your diagram. Describe their behaviour, and what determines this. Then explain how these major parts interact and make up the system as a whole, and so determine the overall system's behaviour. Include in your description the interactions or links between the variables within loops and between loops.

Ans:

1. Number of Sales Representatives: This variable shows how many salespeople work for the organisation. It has a direct impact on order rate; greater order rates are typically associated with more reps.

2. Order per Rep: The typical quantity of orders that every sales representative completes. It influences the company's overall order rate.

3. Order Capacity: The most orders that an organisation can process in a specific amount of time. It affects the demand on production as well as the backlog of orders.

4. Capacity Utilisation as a Percentage: Shows how much of the order capacity is being utilised. Increased production pressure is frequently correlated with higher utilisation.

5. Order Fill Rate: The proportion of successfully completed orders. Both capacity and production pressure have an impact on it.

6. Order Backlog: This is the total number of orders that have not yet been completed. It immediately impacts the demand on production and is impacted by the order rate.

7. manufacturing Strain: Indicates how much the manufacturing process is strained as a result of high order volumes or capacity limitations. It has an impact on both the order backlog and the capacity utilisation percentage.

8. Reported Sales Revenue: The entire amount of money made from sales endeavours. Both the value per order and the average order fill rate have an impact.

9. Average Order Fill Rate: The mean number of orders filled in a given time frame. Both the order fill rate and the time required to reach it have an impact.

The time needed to attain the average order fill rate is known as the "time to average order fill rate." It impacts the average order fill rate in the opposite way.

11. Marketing Department Budget: The amount of money set aside for marketing initiatives. It is affected by the percentage allotted to marketing and the reported sales revenue.

12. Necessary Sales Representative Count: The optimal number of sales representatives required to satisfy demand. Budget for marketing and the difference between needed and current representatives have an impact.

13. Gap: Shows how many sales representatives are needed compared to how many are currently on the job. It affects the hiring and firing rate, which in turn affects the quantity of sales agents.

14. Hire/Fire Rate: The pace at which new representatives are brought on board or at which current representatives are let go. It has an impact on the quantity of sales agents and is impacted by the gap.

15. Time Used to Hire/Fire: The amount of time needed to finish the recruiting or firing procedure. It has the opposite effect on the hire/fire rate.

16. Delivery Delay: The interval of time between an order's scheduled and actual delivery. It immediately impacts the backlog of orders and is impacted by the order fill rate.

17. Delivery Delay Recognised: This indicates when the system has acknowledged a delivery delay. It impacts the acknowledged delivery delay and is impacted by the real delivery delay.

18. Real Delivery Delay: The discrepancy between the scheduled and actual delivery dates. It affects the acknowledged delivery time.

19. Time to Acknowledge Delivery Delay: The amount of time needed to recognise a delivery delay. It is affected in the opposite way by the acknowledged delivery delay.

20. Marketing Percentage: The amount of revenue allotted to advertising. It affects the marketing budget.

21. Order Rate: The total number of orders that the business has ever received. It depends on how many sales reps there are and how many orders each rep takes.

22. Value per Order: The amount of money obtained for every order. It affects the revenue from sales that is recorded.

These factors combine to produce a complicated network in which modifications to one variable can have repercussions on all other variables, affecting resource allocation, production efficiency, and sales performance. The connected pairs show how variables are causally related to one another, creating loops that reinforce and balance the system. Let's dissect each loop and talk about the variables and related actions:

Requirement for Number of Representatives -> Gap -> Hire/Fire Rate -> Number of Sales Representatives -> Order Rate -> Order Rate -> Backlog of Orders -> Pressure on Production -> Percentage Utilisation -> Order Fill Rate -> Average Order Fill Rate -> Reported Sales Revenue -> Marketing Budget

Reinforcing Loop Explanation: As order volume rises, backlog orders accumulate, putting more strain on manufacturing. This pressure raises the capacity utilisation percentage and increases order fill rate. Better average order fill rates are a result of higher order fill rates, which raise reported sales revenue. Increased income enables the marketing budget to be expanded, which in turn permits the hiring of more sales representatives to close the shortfall between the number of required and current reps. Closing the loop, hiring more representatives results in a rise in the number of sales representatives and an increase in the order rate.

Variables and Constants: The starting order rate, order backlog, production pressure, and gap are the loop's constants. Order fill rate, average order fill rate, reported sales revenue, marketing budget, number of sales representatives needed, hire/fire rate, and marketing budget are among the variables.

Potential Measures: Investing in marketing to improve sales income, recruiting more sales reps to manage rising demand, and optimising production capacity are a few possible measures that can be performed to positively reinforce this cycle.

2. Capacity -> Production Pressure -> Order Backlog -> Order Fill Rate -> Real Delivery Delay -> Acknowledged Delivery Delay -> Order Rate -> Ability:

Balanced Loop Explanation: Lower production pressure from increased capacity results in a smaller order backlog. Order fill rates increase when there are fewer backlogged orders, which lowers actual delivery delays. The time it takes to identify delivery delays decreases in tandem with actual delivery delays. Closing the loop, recognising supply delays spurs action to maintain or enhance capacity to meet demand.

Variables and Constants: The order rate and beginning capacity are the loop's constants. Production strain, order backlog, order fill rate, real and recognised delivery delays are some of the variables.

Potential Course of Action: Actions like increasing manufacturing capacity, optimising procedures to cut down on backlog, and enhancing order fulfilment effectiveness can be made to positively reinforce this balancing loop.

Stakeholders can optimise systemic resource allocation, manufacturing efficiency, and sales performance by comprehending these loops and the variables that are linked to them. The given causal loop diagram illustrates a complicated system with numerous interconnected loops that affect different parts of a process or organisation in a reinforcing and balanced manner.

Loop Interactions: The loops depicted in this diagram are related to one another and exert an influence on one another. For instance, the balancing loop B2, which reduces capacity and may restrict the reinforcing effect of R1, can offset the reinforcing loop R1 by increasing the number of late items. This results in a decrease in the intended delivery delay.

Constant Loops: This diagram does not specifically show any constant loops. However, some variables, such the availability of resources, financial limitations, or organisational policies, may be impacted by outside variables or constants that are not displayed.

Direct and Indirect links: The links between the variables are shown in both direct and indirect ways in the diagram. For instance, "Capacity" influences "Percentage Utilisation of Capacity" directly, but through the intermediary variables "Number of Items in Production" and "Number of Late Items," it also influences "Required Number of Items Rate" indirectly.

Potential Course of Action:

R1: The reinforcing loop can be influenced and the quantity of late goods may be decreased by increasing the capacity or changing the target delivery delay.

R2: It is possible to balance the needed number of items rate and the number of late items by controlling the capacity and number of things in production.

B1: Keeping an eye on the capacity utilisation percentage and modifying the intended throughput rate can assist keep capacity and demand in balance.

B2: Managing the quantity of late products can be aided by regulating the acknowledged delivery delay and its effect on capacity.

B3: Maintaining equilibrium between resource allocation and demand can be facilitated by optimising labour hours and capacity utilisation.

Constant Variables: Although not specifically depicted in the graphic, possible constant variables might be things like the maximum capacity that can be used, fixed resource limitations (such a budget or staff), or external market conditions that affect expectations for demand and delivery.

It is crucial to remember that causal loop diagrams are oversimplified depictions of complicated systems; the real dynamics and relationships may be more complex and subject to the effect of a variety of outside variables and presumptions. Proper interpretation and decision-making of the system require a deep comprehension of its context and underlying assumptions.

Loops

Reinforcing Loop R1: This loop illustrates a self-reinforcing cycle in which the number of late items rises and the intended throughput rate rises in response. Consequently, this lowers the intended delivery time and raises capacity. A reduction in the intended delivery delay, however, may result in an increase in the quantity of late products, strengthening the cycle and possibly sending it out of control if improperly handled.

Reinforcing Loop R2: This loop illustrates a cycle in which a rise in capacity causes a rise in the number of items being produced, which raises the rate at which a certain quantity of items is needed. This reverses the initial capacity increase by lowering the quantity of late items and resulting in a drop in the intended throughput rate. By adjusting the available capacity to match the desired throughput rate, this loop may be able to stabilise the system.

Maintaining equilibrium Loop B1: This loop illustrates a cycle in which a rise in the intended throughput rate causes a discrepancy between the intended and actual throughput rates, which in turn causes capacity to increase. A balancing loop that can aid in maintaining a steady level of capacity utilisation is created when capacity increases and its % utilisation falls, counteracting the initial increase in the desired throughput rate.

Loop B2: This loop illustrates a cycle in which a rise in the quantity of delayed items causes the intended delivery delay to drop, hence lowering the acknowledged delivery delay. This creates a balancing loop that can help control delivery delays and capacity based on the quantity of late goods. Consequently, this lowers capacity and may increase the number of late items.

Loop B3 of the balancing loop shows a cycle in which a rise in the percentage of capacity utilised necessitates additional labour hours, which can be met by adding more labour hours. Increased labour hours, on the other hand, result in higher capacity and lower capacity utilisation percentages. This creates a balancing loop that can support the maintenance of a steady level of resource allocation and capacity utilisation.

Impact: This causal loop diagram's variables and loops stand in for several production or delivery system components, including capacity planning, resource allocation, throughput control, and timely delivery. The reinforcing loops (R1, R2) have the ability to produce spiralling or escalating effects, in which a change in one variable can set off a chain reaction of changes in other variables that reinforces itself. If correctly handled, these loops can be advantageous, but if improperly regulated, they can potentially result in instability or uncontrollable growth.

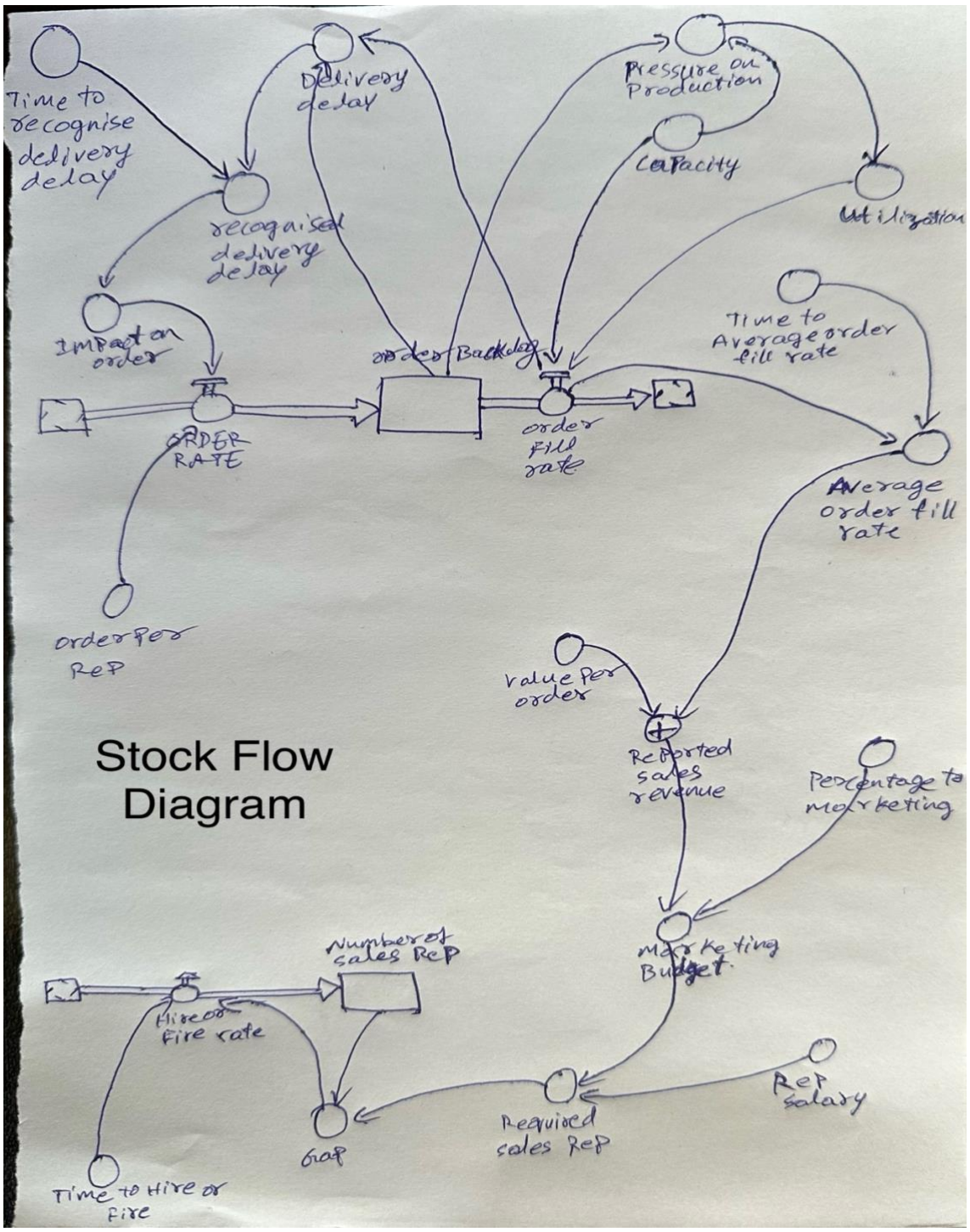
Conversely, the balancing loops (B1, B2, B3) serve as stabilising processes that work to offset the impacts of the reinforcing loops and support the preservation of equilibrium or balance in the system. These loops can be used to control resource allocation, delivery delays, and capacity utilisation based on variables like the percentage of capacity utilised, the number of late goods, and the target throughput rate.

PART B

i) Build up and draw your stock and flow diagram - first do one or more draft versions and include these in the report.

Ans:

Draft Version



ii) Define the equations for the model - so for what determines the rates of flow in and out of stocks, and for those auxiliary variables that need an equation. Check the initial values for your Baseline Model, eg for stocks and the givens or constants, as provided in the study. When correct, as below, please include the list of all equations and values from your iThink model at the equation level.

Ans:

List of Solved Equations:

Order per sales rep

Time to average order fill rate=1

Average Order fill rate - SMTHI(order fill rate - time to average order fill rate).

Price per order. 14,000

Reported sales revenue. Average order fill rate * price per order

Allocated procentage to marketing - 0.16

Marketing dept budget - Reported sales * Allocated percentage to marketing

Montly, salary of sales reps- 4600

Required no of sales seps Marketing dept budget/ monthly salary of sales reps

Grap - Requined no of sales reps - number of sales reps

Time to hire or fire rate - 8

Hire / fire rate= Gap / Time to hire/fire

Order per sales rep -7

Number of sales representatives

Capacity = 85

Required no of Sales rep = Marketing dept budget/ salary of sales rep

Onder Backlog = 85

Number of sales Rep= 4

Actual delivery delay= order backlog / order fill rate.
Pressure on production = order backlog/Capacity

Average order fill rate= SMTHI (order fill rate, Time to average order fill rate)

Order rate = no of sales rep* order per sales rep* impact on orders

Order fill rate= Capacity * percentage utilization of capacity

List of non linear variables

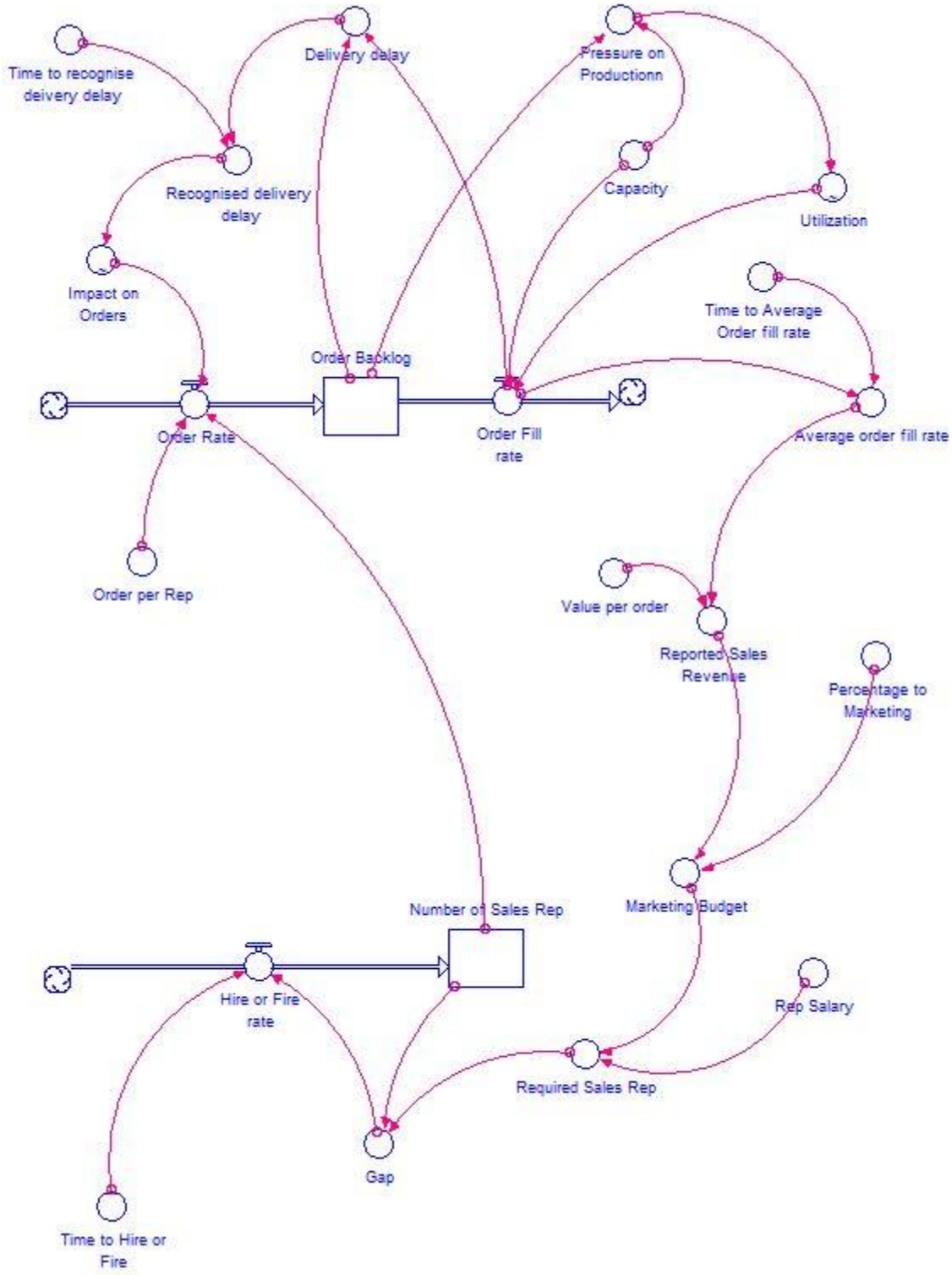
- Recognised delivery delay order rate
- Pressure on production
- Utilisation capacity
- Delivery delay
- Average order fill rate

- Material delay
- Hire/Fire rate
- Information delay

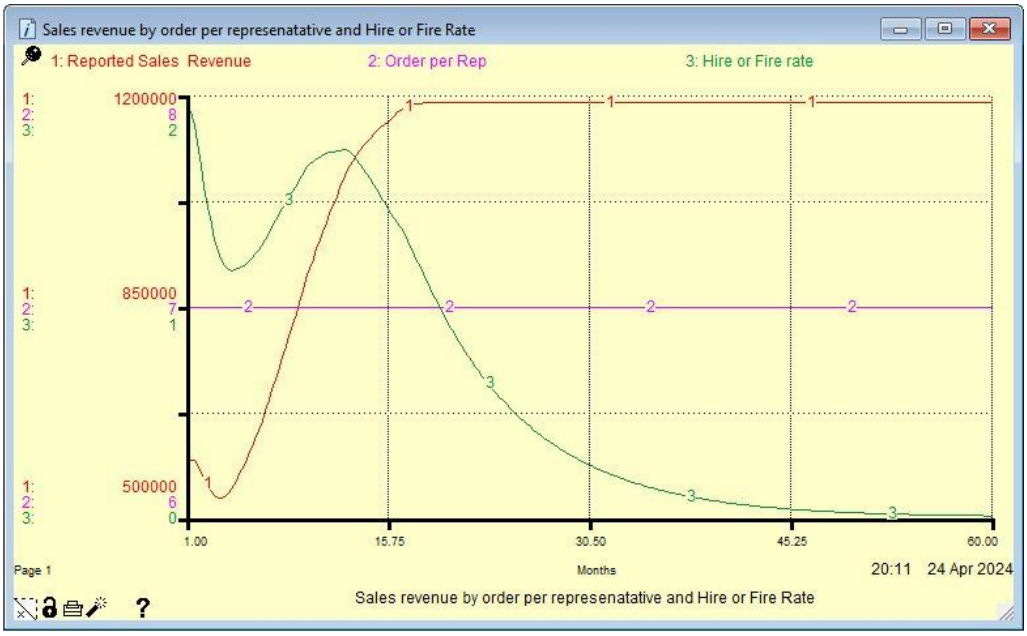
iii) Fully implement the stock and flow model using iThink software, available on the LRC PCs and in the computer labs. You may need to access the software on the LRC computers remotely. Try running the model over a suitable period eg 60 months and create three graphs for a range of key variables, and check if this output confirms your model's validity / a reasonable degree of expected model behaviour. If not, identify the errors and correct them, so it runs successfully. Create table output also for key variable behaviour over the period. Please include the three graphs and table output in your report.

Ans:

Stock Flow Diagram:



Graph 1:



Graph 2:

Graph 3:

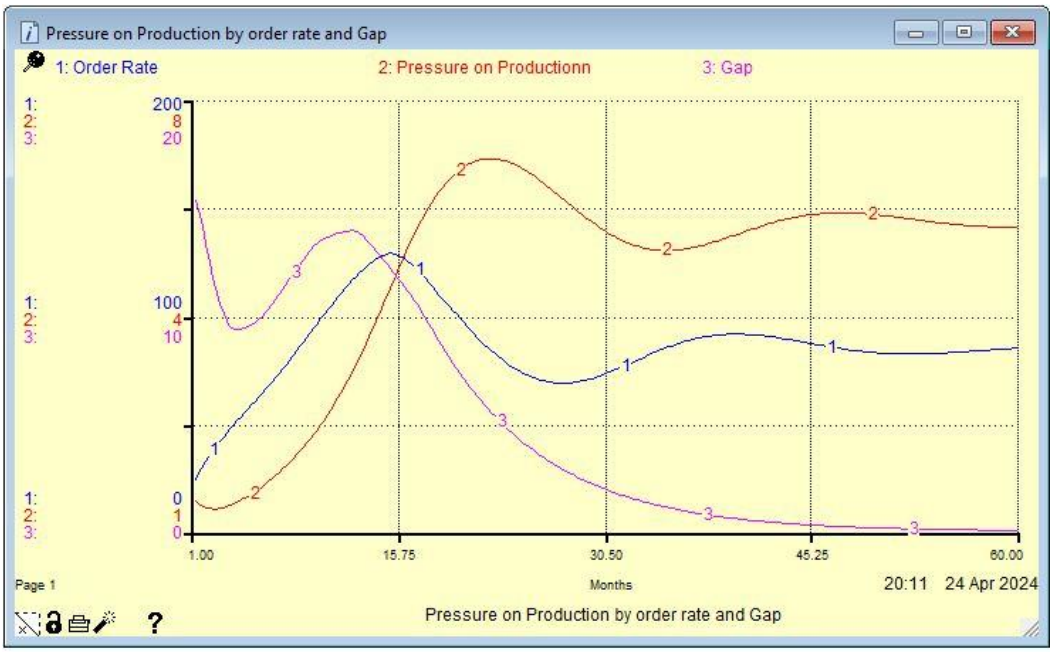

Pressure on Production by order rate and Gap
1: Order Rate
2: Pressure on Productionn
3: Gap
200
8
20
100
4
10
0
1
0
1.00
15.75
30.50
45.25
60.00
Months
Page 1
20:11 24 Apr 2024
Pressure on Production by order rate and Gap

Table Output:

Months	Impact on O	Time to recog	Utilization	Value per ord	Marketing Bu	Hire or Fire ra	Percentage to	Pressure on F	Time to Hire (	Time to Aver	Order Fill rate	Average orde	Rep Salary	Reported Sal
Initial	0.87	6.00	0.50	14,000.00	89,250.00		0.15	1.00	8.00	1.00		42.50	4,600.00	595,000.00
1	0.87	6.00	0.43	14,000.00	84,454.13	1.81	0.15	0.87	8.00	1.00	39.73	40.22	4,600.00	563,027.51
2	0.87	6.00	0.45	14,000.00	79,864.30	1.44	0.15	0.89	8.00	1.00	36.98	38.03	4,600.00	532,428.67
3	0.87	6.00	0.49	14,000.00	82,000.89	1.21	0.15	0.98	8.00	1.00	39.19	39.05	4,600.00	546,672.59
4	0.87	6.00	0.54	14,000.00	88,526.14	1.18	0.15	1.11	8.00	1.00	43.26	42.16	4,600.00	590,174.28
5	0.87	6.00	0.59	14,000.00	96,695.07	1.22	0.15	1.28	8.00	1.00	47.44	46.05	4,600.00	644,633.78
6	0.86	6.00	0.66	14,000.00	106,641.22	1.31	0.15	1.47	8.00	1.00	52.49	50.78	4,600.00	710,941.46
7	0.86	6.00	0.72	14,000.00	117,545.01	1.42	0.15	1.69	8.00	1.00	57.94	55.97	4,600.00	783,633.43
8	0.85	6.00	0.79	14,000.00	128,550.70	1.53	0.15	1.94	8.00	1.00	63.15	61.21	4,600.00	857,004.64
9	0.84	6.00	0.83	14,000.00	139,438.24	1.64	0.15	2.24	8.00	1.00	68.44	66.40	4,600.00	929,588.25
10	0.83	6.00	0.88	14,000.00	148,492.24	1.71	0.15	2.59	8.00	1.00	72.35	70.71	4,600.00	989,948.25
11	0.81	6.00	0.93	14,000.00	157,085.16	1.73	0.15	2.99	8.00	1.00	76.34	74.80	4,600.00	1,047,234.37
12	0.79	6.00	0.95	14,000.00	164,091.03	1.74	0.15	3.44	8.00	1.00	79.54	78.14	4,600.00	1,093,940.17
13	0.76	6.00	0.97	14,000.00	168,641.38	1.68	0.15	3.94	8.00	1.00	81.16	80.31	4,600.00	1,124,275.88
14	0.73	6.00	0.98	14,000.00	172,135.14	1.59	0.15	4.47	8.00	1.00	82.65	81.97	4,600.00	1,147,567.63
15	0.68	6.00	1.00	14,000.00	175,103.72	1.48	0.15	4.99	8.00	1.00	83.88	83.38	4,600.00	1,167,358.16
16	0.62	6.00	1.00	14,000.00	177,419.78	1.38	0.15	5.46	8.00	1.00	84.99	84.49	4,600.00	1,182,798.50
17	0.55	6.00	1.00	14,000.00	178,158.21	1.26	0.15	5.86	8.00	1.00	85.00	84.84	4,600.00	1,187,721.40
18	0.50	6.00	1.00	14,000.00	178,391.86	1.12	0.15	6.16	8.00	1.00	85.00	84.95	4,600.00	1,189,279.04
19	0.45	6.00	1.00	14,000.00	178,465.78	0.99	0.15	6.38	8.00	1.00	85.00	84.98	4,600.00	1,189,771.88
20	0.41	6.00	1.00	14,000.00	178,489.17	0.87	0.15	6.51	8.00	1.00	85.00	84.99	4,600.00	1,189,927.82
21	0.37	6.00	1.00	14,000.00	178,496.57	0.77	0.15	6.56	8.00	1.00	85.00	85.00	4,600.00	1,189,977.16
22	0.34	6.00	1.00	14,000.00	178,498.92	0.68	0.15	6.54	8.00	1.00	85.00	85.00	4,600.00	1,189,992.77
23	0.31	6.00	1.00	14,000.00	178,499.66	0.60	0.15	6.46	8.00	1.00	85.00	85.00	4,600.00	1,189,997.71
24	0.30	6.00	1.00	14,000.00	178,499.89	0.53	0.15	6.33	8.00	1.00	85.00	85.00	4,600.00	1,189,999.28
25	0.28	6.00	1.00	14,000.00	178,499.97	0.46	0.15	6.17	8.00	1.00	85.00	85.00	4,600.00	1,189,999.77
26	0.28	6.00	1.00	14,000.00	178,499.99	0.41	0.15	5.99	8.00	1.00	85.00	85.00	4,600.00	1,189,999.93
27	0.27	6.00	1.00	14,000.00	178,500.00	0.36	0.15	5.80	8.00	1.00	85.00	85.00	4,600.00	1,189,999.98
28	0.28	6.00	1.00	14,000.00	178,500.00	0.32	0.15	5.61	8.00	1.00	85.00	85.00	4,600.00	1,189,999.99
29	0.28	6.00	1.00	14,000.00	178,500.00	0.28	0.15	5.45	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
30	0.29	6.00	1.00	14,000.00	178,500.00	0.25	0.15	5.31	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
31	0.30	6.00	1.00	14,000.00	178,500.00	0.22	0.15	5.20	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
32	0.31	6.00	1.00	14,000.00	178,500.00	0.19	0.15	5.12	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
33	0.32	6.00	1.00	14,000.00	178,500.00	0.17	0.15	5.08	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
34	0.32	6.00	1.00	14,000.00	178,500.00	0.15	0.15	5.07	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
35	0.33	6.00	1.00	14,000.00	178,500.00	0.13	0.15	5.09	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
36	0.34	6.00	1.00	14,000.00	178,500.00	0.11	0.15	5.13	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
37	0.34	6.00	1.00	14,000.00	178,500.00	0.10	0.15	5.19	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
38	0.34	6.00	1.00	14,000.00	178,500.00	0.09	0.15	5.26	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
39	0.34	6.00	1.00	14,000.00	178,500.00	0.08	0.15	5.34	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
40	0.34	6.00	1.00	14,000.00	178,500.00	0.07	0.15	5.42	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
41	0.34	6.00	1.00	14,000.00	178,500.00	0.06	0.15	5.49	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
42	0.33	6.00	1.00	14,000.00	178,500.00	0.05	0.15	5.55	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
43	0.33	6.00	1.00	14,000.00	178,500.00	0.05	0.15	5.60	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
44	0.32	6.00	1.00	14,000.00	178,500.00	0.04	0.15	5.64	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
45	0.32	6.00	1.00	14,000.00	178,500.00	0.04	0.15	5.67	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
46	0.32	6.00	1.00	14,000.00	178,500.00	0.03	0.15	5.68	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
47	0.31	6.00	1.00	14,000.00	178,500.00	0.03	0.15	5.68	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
48	0.31	6.00	1.00	14,000.00	178,500.00	0.03	0.15	5.67	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
49	0.31	6.00	1.00	14,000.00	178,500.00	0.02	0.15	5.65	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
50	0.31	6.00	1.00	14,000.00	178,500.00	0.02	0.15	5.62	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
51	0.30	6.00	1.00	14,000.00	178,500.00	0.02	0.15	5.59	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
52	0.30	6.00	1.00	14,000.00	178,500.00	0.02	0.15	5.56	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
53	0.31	6.00	1.00	14,000.00	178,500.00	0.01	0.15	5.53	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
54	0.31	6.00	1.00	14,000.00	178,500.00	0.01	0.15	5.51	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
55	0.31	6.00	1.00	14,000.00	178,500.00	0.01	0.15	5.48	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
56	0.31	6.00	1.00	14,000.00	178,500.00	0.01	0.15	5.47	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
57	0.31	6.00	1.00	14,000.00	178,500.00	0.01	0.15	5.46	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
58	0.31	6.00	1.00	14,000.00	178,500.00	0.01	0.15	5.45	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00
59	0.31	6.00	1.00	14,000.00	178,500.00	0.01	0.15	5.45	8.00	1.00	85.00	85.00	4,600.00	1,190,000.00

iv) Describe fully the main parts of your stock and flow model and the interactions between the parts. Explain your equations, and this includes how you have modelled both information smoothing and non-linear relationship.

Ans:

Examining Special Patterns in the Interactions between the Stock and Flow Models:-

A stock and flow model can be used by Distinctive Designs to obtain important insights into how their business is run. The interplay between multiple factors that affect the growth trajectory of various aspects is captured by this model.

Stocks:

Backlog of Orders (OB): Represents the number of unfulfilled customer orders. A high backlog can lead to longer wait times and lower customer satisfaction.

Number of Sales Representatives (NSR): Represents the size of the sales force. A larger sales force can lead to more orders but may also incur higher costs.

An extensive analysis of the main stocks, flows, and their interactions is provided below:

Stock/Inventory:

The collection of unmet client requests is known as the backlog of orders (OB). Think of an order container as a symbol for orders that are still pending fulfilment. The number of completed orders is subtracted from the total number of orders in this container to determine the order fill rate. Order Rate, on the other hand, just counts the number of new orders. An increasing backlog of clients awaiting their purchases is indicated by a high OB. This might put strain on the ability to produce goods and result in lengthier wait periods.

NSR Sales Representatives:

This represents the size of the sales force. The Net Staffing Rate (NSR) has a positive association with the hiring rate and a negative correlation with the firing rate. An elevated net promoter ratio (NSR) indicates the possibility of boosting order volume through enhanced customer interaction. However, there is a chance that returns will decrease as soon as the company hits market saturation in the intended area.

Customers' perceptions of wait times have an impact on this company's stock, as evidenced by the Acknowledged Delivery Delay (RDD). A modified version of the actual delivery delay that accounts for its influence is called the Time to Recognise Delivery Delay (TRDD). Customers' perceptions of major delays might be reflected in high RDD (Relative Delay Discomfort) scores, which can affect their satisfaction and subsequent purchasing decisions.

Change/Flow Rate:

Rate of Order (OR): The mean monthly number of new orders placed is shown by this measure. It is impacted by two things:

NSR Sales Representatives: Even though multiple salespeople may make orders, there may not always be a precisely linear relationship between the number of salespeople and the quantity of orders. There's a chance of seeing declining results when the sales force expands and reaches its maximum potential in the target market for qualifying leads.

Average Rep Order (not shown):

The average order value that a salesperson generates is represented by this constant value.

One way to calculate the percentage of client orders that are successfully completed is to utilise the Order Fill Rate (OFR) indicator. The volume of completed orders every month is represented by this flow. This is determined by the company's secret production capacity and the Utilisation Rate, which measures how well production uses its capacity. Orders are processed more quickly with a greater Order Fulfilment Rate (OFR), which lowers the Order Backlog and shortens client wait times.

Rate of Hiring (HR): This flow is a representation of the new salesmen hired each month. It is based on the difference between the current NSR and the required number of sales representatives, which is established by sales revenue projections. A positive gap suggests that hiring is required since it shows that more salespeople are needed to handle the high volume of sales.

The speed at which a firearm may fire bullets. This flow shows how many salespeople leave each month. The hiring rate and gap are two related factors that affect the result together. When there is a negative gap, or too many salespeople, termination is required to optimise costs and manpower levels.

Delivery Delay (DD):

This procedure shows how long it takes to generate and deliver an order. It is impacted by two things:

Backlog of Orders (OB): Extended wait times for completing individual orders could arise from a significant backlog that strains production capabilities.

One way to calculate the percentage of client orders that are successfully completed is to utilise the Order Fill Rate (OFR) indicator. It is possible to shorten the time that customers must wait for new orders by expediting the processing of current ones.

Smoothing of Information:

The Ordinance Fill Rate on Average (AOFR) is the average rate at which ordinances are filled. This shows the average Order Fill Rate for a given month. By balancing out variations in the real Output Gap Financial Ratio (OFR), it gives the model's computations a more consistent figure.

Innate Connections:

The proportion of sales reps to orders. The quantity of salespeople and the order rate have a non-linear connection. Orders will probably increase if there are more salesmen on the team, but the increase might not be proportionate. The returns a firm earns may decline once it reaches a point when it has saturated the market in the industry it is targeting.

Equation between Customer Orders and Delivery Delays:

Orders from customers are not linearly affected by delivery delays. Orders are not likely to be significantly impacted by a little increase in delivery delays. However, if there is a long delay, customers can become impatient and look for other suppliers, which could result in a significant drop in orders.

Critical Links:

Order backlog and rate:

The Order Rate has a direct impact on the Order Backlog. As more orders are received, the amount of pending orders grows, placing pressure on the manufacturing process to keep up speed. A significant backlog could result in longer wait times and possibly worse customer satisfaction.

Order fulfilment rate, backlog of orders, and delivery delays: The order fill rate has a direct impact on the backlog of orders and delivery delays. Shorter wait times for clients result directly from faster task completion, which lowers the quantity of unfinished work. An effective production process with a high Order Fill Rate is required to maintain an adequate backlog and guarantee customer satisfaction.

Order backlog and delivery delay:

The order backlog and order fill rate taken together determine the delivery delay. Long wait times for clients may arise from a large backlog of unfulfilled orders combined with a low order fulfilment rate.